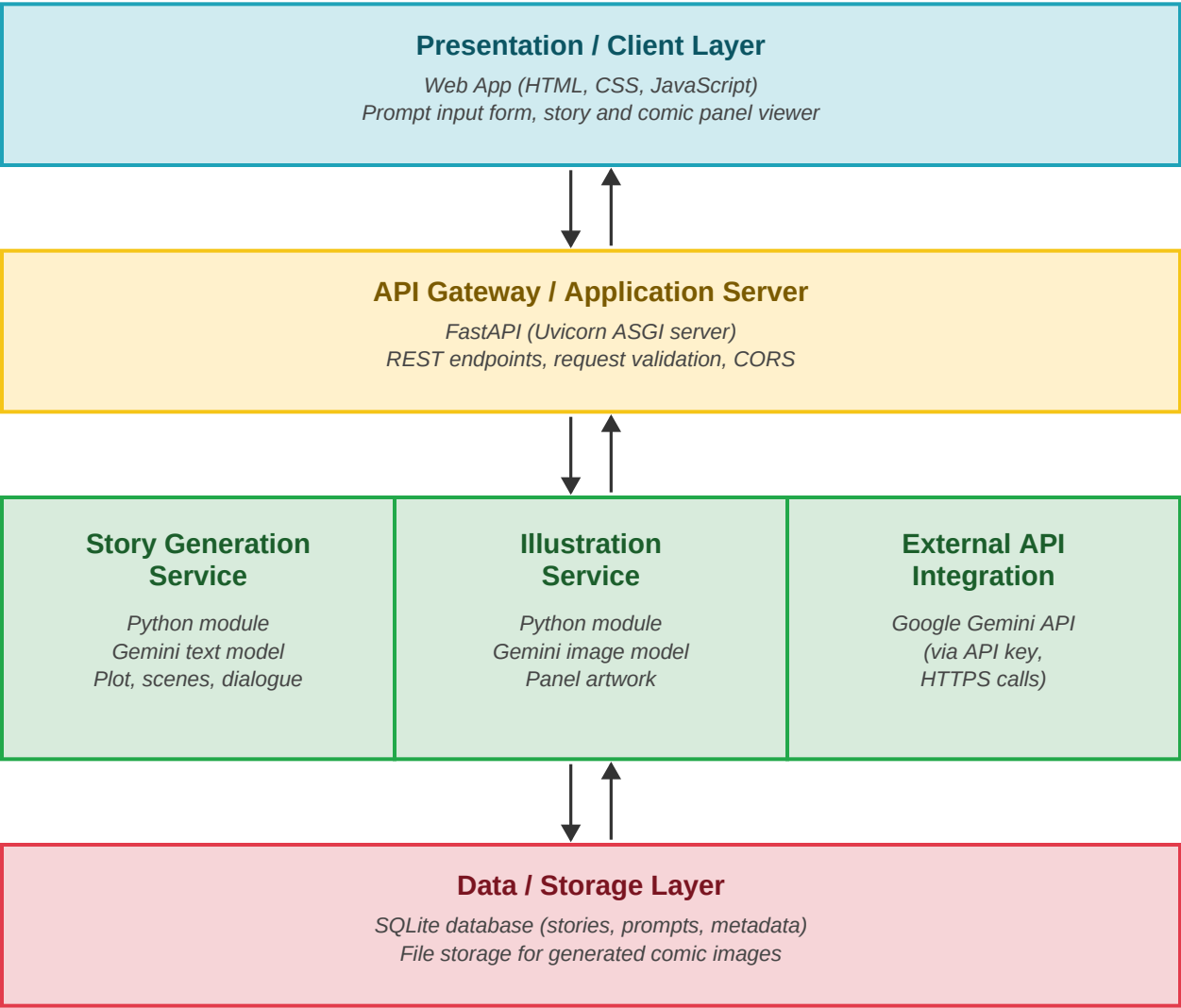


Solution Architecture

Date	29 September 2026
Team ID	UNM123APEXLAB
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:



Flow: User prompt -> FastAPI -> Story service -> Illustration service -> Gemini API -> stored comic -> displayed in browser.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web interface where the user enters a story idea or prompt, chooses style/genre, and views the generated comic story with illustrated panels.	HTML, CSS, JavaScript
API Gateway	Single entry point for the client. Receives requests, validates input, routes them to the story and image services, and returns JSON/image responses.	FastAPI, Uvicorn, Pydantic
Microservices (Story + Illustration)	Story service turns the prompt into a structured comic script (scenes, captions, dialogue). Illustration service generates a panel image for each scene.	Python, Gemini text model, Gemini image model
External API Integration	Secure connection to Google's generative AI platform that powers both story and image generation.	Google Gemini API, Python SDK, API key in .env
Database / Storage	Stores prompts, generated stories, panel metadata and the generated images so comics can be reloaded and downloaded.	SQLite, local file storage